

PORTFOLIO CASE STUDY · CUSTOMER DATA

Customer Behavior Analysis

From 3,900 purchases to evidence-led business decisions.

Python

SQL

Power BI

Made with GAMMA



PROJECT FRAME

A 3,900-row dataset turns into a recruiter-ready case study

- 1 Business lens:** understand revenue mix, customer loyalty, subscription behavior, discount exposure and delivery choice
- 2 Toolkit:** SQL aggregation, conditional logic, window functions and customer segmentation
- 3 Output:** a visual analysis designed to move from evidence to action, not just show a dashboard

The headline is scale: \$233,081 in observed revenue

\$233k

OBSERVED REVENUE

3,900

purchases

\$59.76

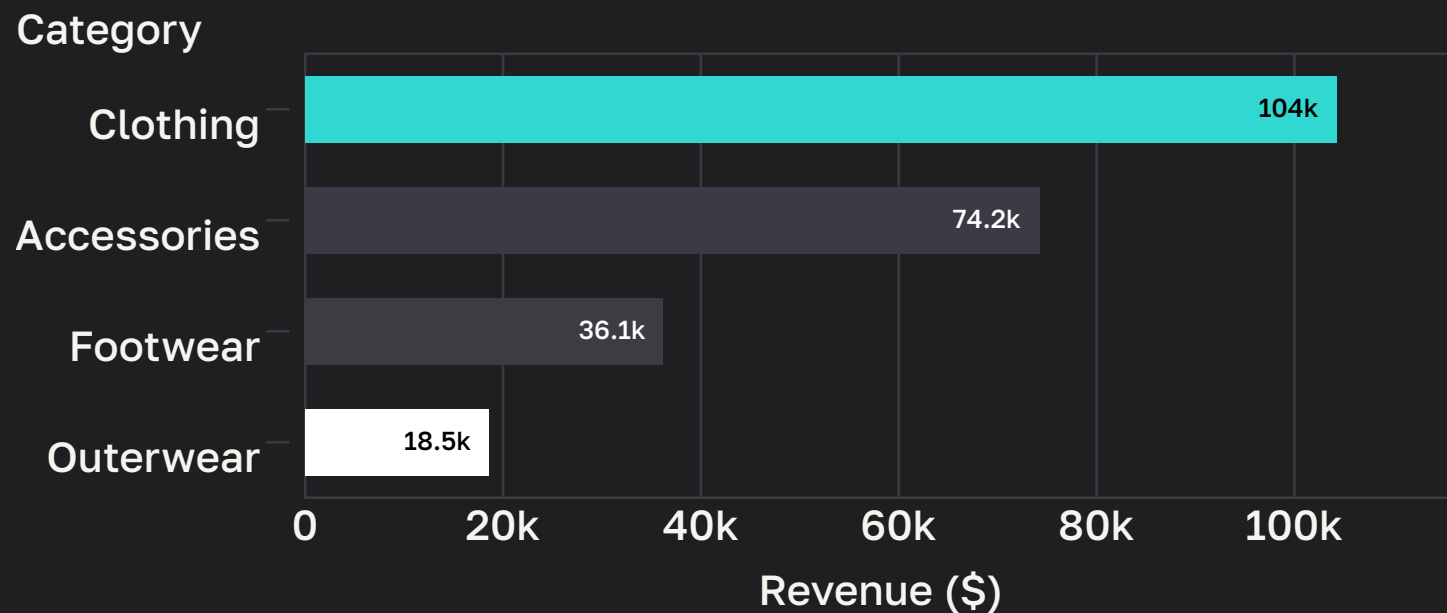
average purchase amount

Mix already has a shape

Clothing leads at \$104,264, followed by **Accessories** at \$74,200.

A CLEAN, AUDITABLE BASE FOR THE ANALYSIS.

Clothing drives the business, but Accessories is the second engine



45%

Clothing's share of observed revenue.

32%

Accessories' share.



CLOTHING



ACCESSORIES

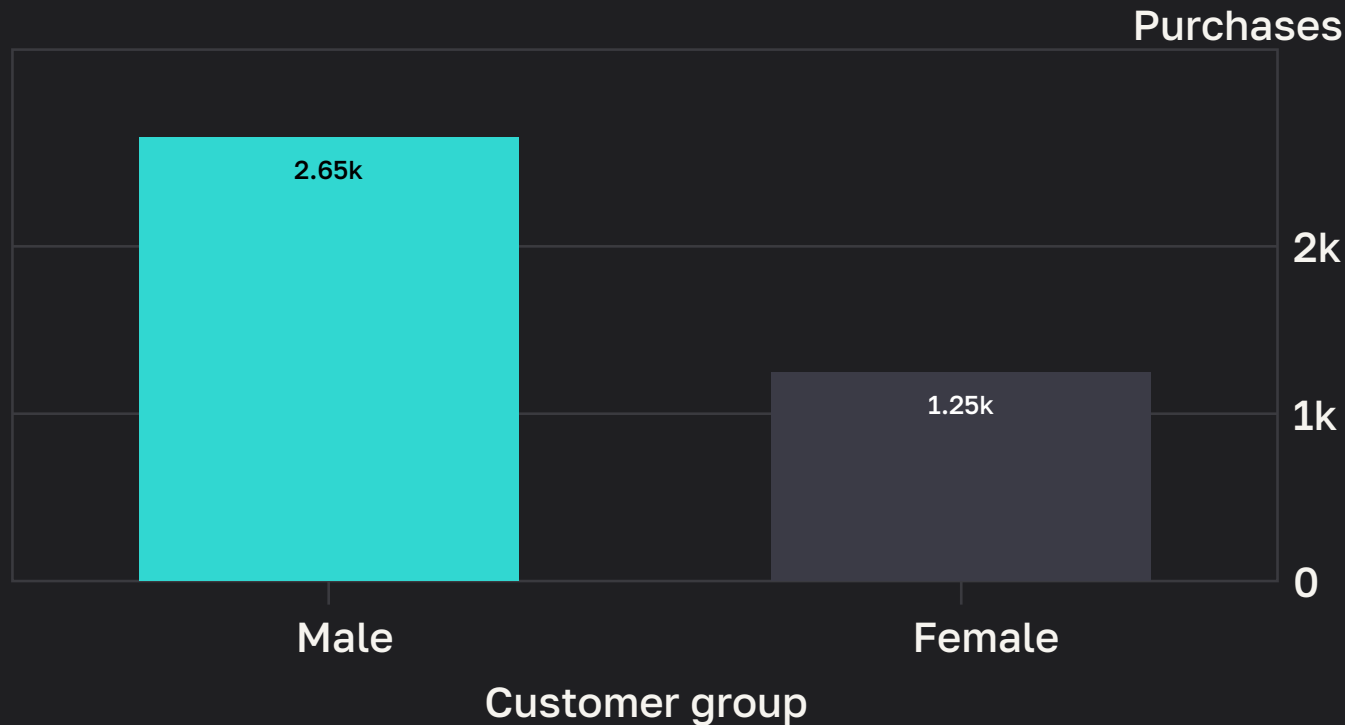


FOOTWEAR



OUTERWEAR

The gender split is large, but it is a mix signal, not a value judgment



2,652 vs 1,248

Male customers contribute \$157,890;
female customers contribute \$75,191.

AVERAGE PURCHASE AMOUNT IS NOT THE
STORY. THE OBSERVED DIFFERENCE IS
PRIMARILY CUSTOMER MIX.

Subscription is not yet translating into higher spend

AVERAGE PURCHASE AMOUNT

\$59.49

Subscribers

\$59.87

Non-subscribers

Volume explains the revenue gap

Subscribers represent **1,053 purchases**. Non-subscribers represent **2,847**.

Test subscription on retention, frequency or margin rather than assuming it lifts ticket size.

Product demand is broad, so ratings help prioritize the next question

TOP AVERAGE-RATED PRODUCTS

3.86



Gloves

3.84



Sandals

3.82



Boots

The next cut is category-specific

Clothing: Blouse and Pants

Accessories: Jewelry

Footwear: Sandals

Outerwear: Jacket

PAIR DEMAND WITH RATING BEFORE CHOOSING THE NEXT PRODUCT QUESTION.

Discounts are common, while shipping changes the ticket only slightly

43%

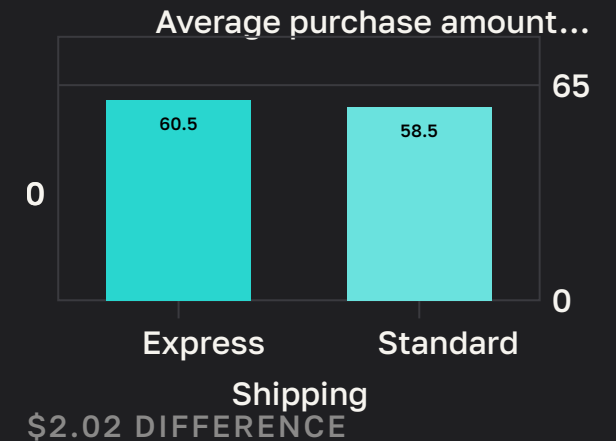
of purchases include a discount

1,677 OF 3,900 PURCHASES

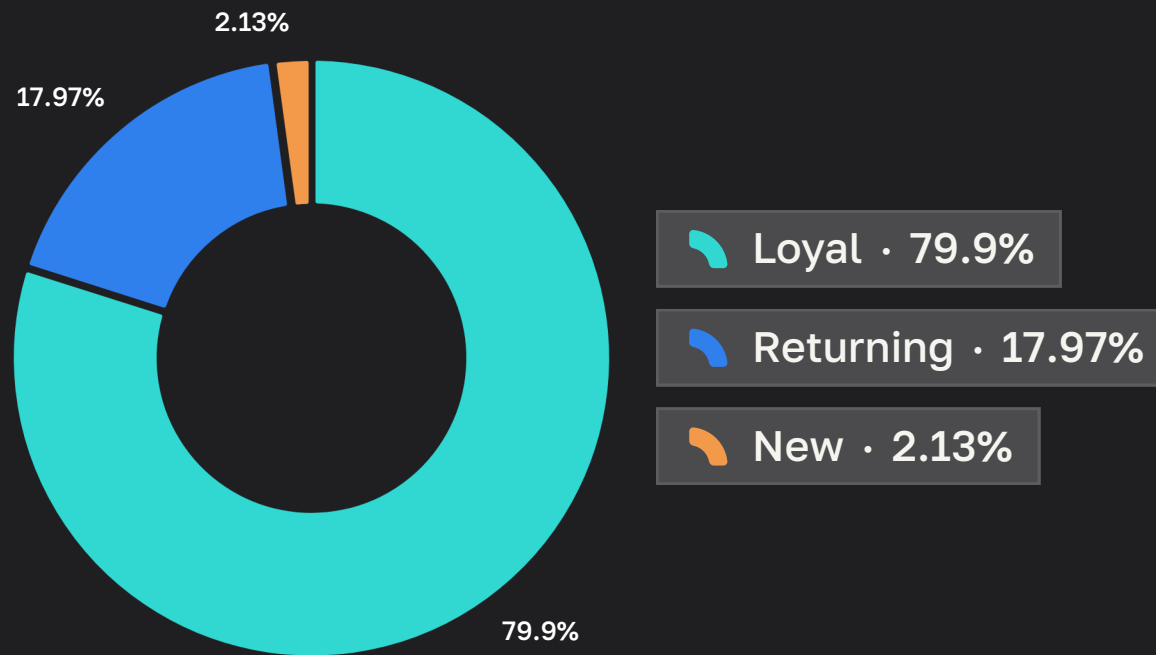
Highest incidence

Hat: 50.0%

Sneakers: 49.7%



The customer base is already overwhelmingly Loyal



3,116 Loyal

Using previous purchases: New = 1 purchase, Returning = 2–10, Loyal = more than 10.

Protecting repeat behavior is as important as acquisition.

The portfolio move: connect the query to the decision

TEST NEXT

Subscription offers

Tie them to frequency.

GUARDRAILS

Category discounts

Set category-specific limits.

RETENTION

High-purchase customers

Sharpen the messaging.

What this project demonstrates: turning raw customer data into clear metrics, structured SQL, visual storytelling and business recommendations.

AN ANALYSIS IS STRONGEST WHEN EVERY CHART ENDS WITH A DECISION, A TEST OR A NEXT QUESTION.